

Multi-Modal Interactive Engagement System: Expression Engine & Creative Activities

1. Nanny Unit: Emotional Expression Engine with Mirror Neuron Integration

A. Animated Expression System Architecture

python

```
class EmotionalExpressionEngine:
    def __init__(self, hardware_interface):
        self.hardware = hardware_interface
        self.expression_states = self.initialize_expressions()
        self.mirror_neuron_engine = MirrorNeuronEngine()
        self.expression_history = []
        self.current_mood = 'neutral'

    def initialize_expressions(self):
        """
        Define comprehensive emotional expressions with animation properties
        """
        return {
            'happy': {
                'eyes': {
                    'shape': 'wide_open',
                    'eyebrows': 'raised_curved',
                    'pupils': 'dilated',
                    'sparkle': True,
                    'blink_rate': 'fast_happy',
                    'animation': 'gentle_bounce'
                },
                'mouth': {
                    'shape': 'wide_smile',
                    'curve': 'upward_strong',
                    'teeth': 'visible',
                    'tongue': 'peeking',
                    'animation': 'smile_grow'
                },
                'accessories': ['sparkles', 'glow', 'confetti'],
                'duration': 3.0,
                'transition': 'smooth'
            },
            'sad': {
                'eyes': {
                    'shape': 'drooping',
                    'eyebrows': 'inner_raised',
                    'pupils': 'constricted',
                    'teardrop': {
                        'size': 'medium',
                        'flow': 'slow_drip',
                        'count': 1
                    },
                    'blink_rate': 'slow_heavy',
                    'animation': 'gentle_droop'
                }
            }
        }
```

```

    'mouth': {
        'shape': 'small_frown',
        'curve': 'downward',
        'quiver': True,
        'animation': 'tremble'
    },
    'accessories': ['raindrops', 'fade_out'],
    'duration': 2.5,
    'transition': 'slow_fade'
},

'excited': {
    'eyes': {
        'shape': 'wide_stars',
        'eyebrows': 'high_arched',
        'pupils': 'stars',
        'sparkle': 'intense',
        'blink_rate': 'rapid',
        'animation': 'sparkle_burst'
    },
    'mouth': {
        'shape': 'open_smile',
        'curve': 'upward_extreme',
        'teeth': 'visible',
        'tongue': 'peeking',
        'animation': 'bounce_chatter'
    },
    'accessories': ['fireworks', 'sparkle_trail', 'bounce_effect'],
    'duration': 4.0,
    'transition': 'bounce_in'
},

'worried': {
    'eyes': {
        'shape': 'narrow_focused',
        'eyebrows': 'furrowed',
        'pupils': 'small_focused',
        'glance': 'side_to_side',
        'blink_rate': 'rapid_nervous',
        'animation': 'worried_glance'
    },
    'mouth': {
        'shape': 'tight_line',
        'curve': 'slight_down',
        'biting_lip': True,
        'animation': 'nervous_twitch'
    },
    'accessories': ['sweat_drop', 'pulse_glow'],
    'duration': 2.0,
    'transition': 'tense_appear'
},

'thinking': {
    'eyes': {
        'shape': 'squinted',
        'eyebrows': 'one_raised',
        'pupils': 'upward_gaze',
        'thought_bubble': True,
        'blink_rate': 'thoughtful_slow',
        'animation': 'processing_pulse'
    },
    'mouth': {
        'shape': 'pursed',
        'curve': 'neutral',

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```

        'bite': 'lip',
        'animation': 'thoughtful_move'
    },
    'accessories': ['thought_bubbles', 'processing_dots',
'lightbulb'],
    'duration': 3.5,
    'transition': 'gradual_appear'
},

'robot_happy_dance': {
    'eyes': {
        'shape': 'pixel_hearts',
        'eyebrows': 'bouncing',
        'pupils': 'dancing',
        'blink_rate': 'disco',
        'animation': 'dance_sync'
    },
    'mouth': {
        'shape': 'pixel_smile',
        'curve': 'bouncing',
        'pixels': 'dancing',
        'animation': 'beat_sync'
    },
    'accessories': ['disco_lights', 'robot_arms', 'beat_pulse'],
    'duration': 5.0,
    'transition': 'robot_boot'
}
}

```

B. Mirror Neuron Engine for Emotional Synchronization

python

```

class MirrorNeuronEngine:
    def __init__(self):
        self.emotional_mirroring = {
            'direct_mirror': True, # Directly mirror user's emotion
            'complementary_response': True, # Respond with complementary
emotion
            'emotional_scaffolding': True, # Guide toward healthier emotional
state
            'cultural_adjustments': self.load_cultural_emotion_rules()
        }
        self.mirror_history = []

    def determine_appropriate_response(self, user_emotion, user_profile):
        """
        Determine whether to mirror, complement, or scaffold user emotion
        """
        # Get emotion characteristics
        emotion_data = self.analyze_emotion(user_emotion)

        # Decision matrix for response type
        if user_profile.get('cognitive_condition') in ['autism', 'adhd']:
            # Clear, exaggerated mirroring for neurodivergent users
            return {
                'type': 'exaggerated_mirror',
                'intensity': emotion_data['intensity'] * 1.3,
                'duration': emotion_data['duration'] * 1.2
            }

        elif user_profile.get('age', 30) > 70:
            # Gentle, clear responses for elderly

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        return {
            'type': 'gentle_mirror',
            'intensity': min(emotion_data['intensity'] * 0.8, 0.7),
            'duration': emotion_data['duration'] * 1.5
        }

    elif emotion_data['valence'] == 'negative' and emotion_data['intensity']
> 0.7:
        # High negative emotion: scaffold toward neutral/positive
        return {
            'type': 'emotional_scaffolding',
            'target_emotion': self.select_scaffolding_target(user_emotion),
            'transition_rate': 0.3
        }

    elif emotion_data['valence'] == 'positive':
        # Positive emotion: mirror with enthusiasm
        return {
            'type': 'enhanced_mirror',
            'intensity': emotion_data['intensity'] * 1.1,
            'duration': emotion_data['duration']
        }

    else:
        # Default: direct mirror
        return {
            'type': 'direct_mirror',
            'intensity': emotion_data['intensity'],
            'duration': emotion_data['duration']
        }

    def emotional_scaffolding(self, current_emotion, target_emotion,
user_profile):
        """
        Gradually guide user from negative to more positive emotional state
        """
        scaffolding_paths = {
            'anger': ['frustration', 'determination', 'focus'],
            'sadness': ['melancholy', 'reflection', 'acceptance', 'hope'],
            'fear': ['caution', 'awareness', 'curiosity'],
            'frustration': ['challenge', 'determination', 'accomplishment']
        }

        path = scaffolding_paths.get(current_emotion, [current_emotion,
'neutral'])

        return {
            'current_emotion': current_emotion,
            'target_path': path,
            'transition_steps': len(path),
            'step_duration': 2.0, # seconds per emotional transition
            'visual_transitions': self.generate_scaffolding_animations(path)
        }

    def generate_scaffolding_animations(self, emotion_path):
        """
        Generate smooth transitions between emotional states
        """
        animations = []

        for i in range(len(emotion_path) - 1):
            from_emotion = emotion_path[i]
            to_emotion = emotion_path[i + 1]

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        animations.append({
            'from': from_emotion,
            'to': to_emotion,
            'transition_type': 'morph',
            'duration': 1.5,
            'visual_cues': [
                'color_gradient',
                'shape_morph',
                'expression_blend'
            ]
        })

    return animations

```

C. Real-Time Expression Display System

python

```

class ExpressionDisplay:
    def __init__(self):
        self.canvas = None
        self.animation_queue = []
        self.active_effects = []
        self.render_engine = ExpressionRenderEngine()

    def initialize_display(self, display_type='rpi_screen'):
        """
        Initialize display based on available hardware
        """
        if display_type == 'rpi_screen':
            self.setup_rpi_framebuffer()
        elif display_type == 'web_interface':
            self.setup_web_canvas()
        elif display_type == 'external_monitor':
            self.setup_hdmi_output()

        # Create expression layers
        self.create_expression_layers()

    def create_expression_layers(self):
        """
        Create layered canvas for complex expressions
        """
        self.layers = {
            'background': Layer(z_index=0, opacity=1.0),
            'face_base': Layer(z_index=1, opacity=1.0),
            'eyes': Layer(z_index=2, opacity=1.0),
            'mouth': Layer(z_index=3, opacity=1.0),
            'eyebrows': Layer(z_index=4, opacity=1.0),
            'accessories': Layer(z_index=5, opacity=0.8),
            'effects': Layer(z_index=6, opacity=0.6),
            'overlay': Layer(z_index=7, opacity=1.0)
        }

    def display_expression(self, expression_name, intensity=1.0, duration=None):
        """
        Display an emotional expression with specified intensity
        """
        expression = self.expression_engine.expression_states[expression_name]

        # Adjust expression based on intensity
        adjusted_expression = self.adjust_intensity(expression, intensity)

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# Render each layer
self.render_face_base(adjusted_expression)
self.render_eyes(adjusted_expression['eyes'])
self.render_mouth(adjusted_expression['mouth'])
self.render_accessories(adjusted_expression.get('accessories', []))

# Apply animations
self.apply_animations(adjusted_expression['animation'])

# Schedule expression duration
if duration:
    self.schedule_expression_change(duration, 'neutral')

def render_eyes(self, eye_config):
    """
    Render animated eyes with emotion-specific properties
    """
    # Eye shapes
    shapes = {
        'wide_open': self.draw_wide_eyes(),
        'drooping': self.draw_droopy_eyes(),
        'squinted': self.draw_squinted_eyes(),
        'pixel_hearts': self.draw_heart_eyes(),
        'wide_stars': self.draw_starry_eyes()
    }

    # Eye animations
    animations = {
        'gentle_bounce': self.animate_bounce(amplitude=2, frequency=0.5),
        'worried_glance': self.animate_glance(rate=0.3, range=20),
        'processing_pulse': self.animate_pulse(speed=0.5, intensity=0.3),
        'sparkle_burst': self.animate_sparkle(count=15, duration=1.0),
        'dance_sync': self.animate_dance(beat_detected=True)
    }

    # Apply shape and animation
    eye_shape = shapes.get(eye_config['shape'], shapes['wide_open'])
    eye_animation = animations.get(eye_config['animation'])

    self.layers['eyes'].draw(eye_shape)
    self.layers['eyes'].animate(eye_animation)

    # Add tears if specified
    if eye_config.get('teardrop'):
        self.render_teardrop(eye_config['teardrop'])

def render_teardrop(self, teardrop_config):
    """
    Render animated teardrops
    """
    tear_properties = {
        'size': teardrop_config.get('size', 'medium'),
        'flow_speed': teardrop_config.get('flow', 'slow_drip'),
        'count': teardrop_config.get('count', 1),
        'transparency': 0.7,
        'reflection': True
    }

    # Create tear animation path
    tear_path = self.calculate_tear_path(tear_properties)

    # Draw tears on effects layer
    for i in range(tear_properties['count']):
        tear = self.draw_teardrop(

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        position=tear_path['start'][i],
        size=tear_properties['size']
    )

    self.layers['effects'].add_object(
        tear,
        animation=self.animate_teardrop_fall(tear_path['path'][i])
    )

def render_accessories(self, accessories):
    """
    Render emotional accessories (sparkles, fireworks, etc.)
    """
    accessory_renderers = {
        'sparkles': self.render_sparkles,
        'fireworks': self.render_fireworks,
        'confetti': self.render_confetti,
        'raindrops': self.render_raindrops,
        'disco_lights': self.render_disco_lights,
        'thought_bubbles': self.render_thought_bubbles,
        'robot_arms': self.render_robot_arms
    }

    for accessory in accessories:
        if accessory in accessory_renderers:
            accessory_renderers[accessory]()

def render_fireworks(self):
    """
    Render celebratory fireworks animation
    """
    fireworks_config = {
        'count': random.randint(5, 15),
        'types': ['burst', 'fountain', 'willow', 'peony'],
        'colors': ['#FF6B6B', '#4ECDC4', '#FFD166', '#06D6A0', '#118AB2'],
        'duration': 3.0,
        'sound_effect': 'fireworks_pop' if self.has_audio else None
    }

    # Create multiple firework bursts
    for i in range(fireworks_config['count']):
        firework_type = random.choice(fireworks_config['types'])
        color = random.choice(fireworks_config['colors'])

        self.create_firework_burst(
            type=firework_type,
            color=color,
            position=self.random_screen_position(),
            delay=i * 0.2
        )

def render_robot_arms(self):
    """
    Render dancing robot arms animation
    """
    arm_config = {
        'segments': 3,
        'joint_angles': [30, -20, 45], # Degrees for each joint
        'dance_pattern': 'robot_disco',
        'beat_sync': True,
        'glow_effect': True
    }

    # Create left and right arms

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left_arm = self.create_robot_arm(
    side='left',
    base_position=(100, 300),
    config=arm_config
)

right_arm = self.create_robot_arm(
    side='right',
    base_position=(700, 300),
    config=arm_config
)

# Animate arms in sync
animation = self.create_robot_dance_animation(
    left_arm=left_arm,
    right_arm=right_arm,
    pattern=arm_config['dance_pattern']
)

self.layers['accessories'].add_animation(animation)

```

D. Web Interface for Expression Display

javascript

```

// Frontend expression display system
class WebExpressionDisplay {
    constructor(canvasId) {
        this.canvas = document.getElementById(canvasId);
        this.ctx = this.canvas.getContext('2d');
        this.layers = {};
        this.currentExpression = 'neutral';
        this.animationLoop = null;

        this.initializeLayers();
        this.setupEventListeners();
    }

    initializeLayers() {
        this.layers = {
            background: new Layer('background', 0),
            face: new Layer('face', 1),
            eyes: new Layer('eyes', 2),
            mouth: new Layer('mouth', 3),
            eyebrows: new Layer('eyebrows', 4),
            accessories: new Layer('accessories', 5),
            effects: new Layer('effects', 6)
        };
    }

    setExpression(expressionName, intensity = 1.0) {
        const expression = EXPRESSION_LIBRARY[expressionName];

        if (!expression) {
            console.error(`Expression ${expressionName} not found`);
            return;
        }

        this.currentExpression = expressionName;

        // Clear previous animations
        this.stopAllAnimations();
    }
}

```



```

        // Apply expression with intensity
        this.applyExpression(expression, intensity);

        // Start animation loop
        this.startAnimationLoop();
    }

    applyExpression(expression, intensity) {
        // Adjust expression parameters based on intensity
        const adjusted = this.adjustIntensity(expression, intensity);

        // Render each component
        this.renderFaceBase(adjusted.face);
        this.renderEyes(adjusted.eyes);
        this.renderMouth(adjusted.mouth);
        this.renderEyebrows(adjusted.eyebrows);

        // Add accessories
        if (adjusted.accessories) {
            this.renderAccessories(adjusted.accessories);
        }

        // Start animations
        this.startComponentAnimations(adjusted.animations);
    }

    renderEyes(eyeConfig) {
        // Draw eye shapes
        const leftEye = this.drawEye(
            'left',
            eyeConfig.shape,
            eyeConfig.position.left
        );

        const rightEye = this.drawEye(
            'right',
            eyeConfig.shape,
            eyeConfig.position.right
        );

        // Apply eye animations
        if (eyeConfig.animation) {
            this.animateEyes(leftEye, rightEye, eyeConfig.animation);
        }

        // Add tears if needed
        if (eyeConfig.teardrop) {
            this.renderTears(eyeConfig.teardrop);
        }

        // Add sparkles
        if (eyeConfig.sparkle) {
            this.addEyeSparkles(leftEye, rightEye);
        }
    }

    renderTears(teardropConfig) {
        // Create tear animation
        const tearAnimation = {
            startY: -20,
            endY: this.canvas.height + 20,
            speed: teardropConfig.speed || 2,
            count: teardropConfig.count || 1,
            size: teardropConfig.size || 'medium'
        };
    }

```

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    };

    // Create tear particles
    for (let i = 0; i < tearAnimation.count; i++) {
        const tear = this.createTearParticle(tearAnimation);
        this.layers.effects.addParticle(tear);
    }
}

// Fireworks effect
renderFireworks(count = 10) {
    for (let i = 0; i < count; i++) {
        setTimeout(() => {
            this.createFireworkBurst(
                this.randomCanvasPosition(),
                this.randomColor()
            );
        }, i * 200); // Staggered start
    }
}

// Robot dance animation
renderRobotDance() {
    const dancePatterns = [
        'robot_wave',
        'robot_shuffle',
        'robot_spin',
        'robot_jump'
    ];

    // Sequence of dance moves
    const danceSequence = [
        { pattern: 'robot_wave', duration: 2000 },
        { pattern: 'robot_shuffle', duration: 1500 },
        { pattern: 'robot_spin', duration: 3000 },
        { pattern: 'robot_jump', duration: 1000 }
    ];

    this.executeDanceSequence(danceSequence);
}

// WebSocket integration for real-time expression updates
class ExpressionWebSocket {
    constructor(serverUrl) {
        this.ws = new WebSocket(serverUrl);
        this.display = new WebExpressionDisplay('marvinFace');

        this.setupWebSocket();
    }

    setupWebSocket() {
        this.ws.onmessage = (event) => {
            const data = JSON.parse(event.data);

            switch (data.type) {
                case 'expression':
                    this.display.setExpression(data.expression, data.intensity);
                    break;

                case 'effect':
                    this.display[`render${data.effect}`]();
                    break;
            }
        };
    }
}

```

```

        case 'dance':
            this.display.renderRobotDance();
            break;

        case 'mirror_emotion':
            this.display.mirrorUserEmotion(data.emotion);
            break;
    }
};
}

sendUserEmotion(emotion) {
    // Send user's detected emotion to server
    this.ws.send(JSON.stringify({
        type: 'user_emotion',
        emotion: emotion,
        timestamp: Date.now()
    }));
}
}

```

2. Interactive Drawing & Coloring Program

A. Accessible Drawing Interface

python

```

class InteractiveDrawingProgram:
    def __init__(self, user_profile):
        self.user_profile = user_profile
        self.canvas_config = self.initialize_canvas(user_profile)
        self.tools = self.initialize_tools(user_profile)
        self.coloring_books = self.load_coloring_books(user_profile)
        self.ai_assistant = DrawingAIAssistant()
        self.accessibility_features = self.setup_accessibility(user_profile)

    def initialize_canvas(self, user_profile):
        """
        Create canvas optimized for user's abilities
        """
        config = {
            'size': self.determine_canvas_size(user_profile),
            'background': self.select_background(user_profile),
            'grid': self.setup_grid(user_profile),
            'zoom': self.setup_zoom_controls(user_profile),
            'undo_redo': self.setup_undo_redo(user_profile)
        }

        # Adjust for cognitive conditions
        if user_profile.get('cognitive_condition'):
            condition = user_profile['cognitive_condition']
            if condition in ['dementia', 'alzheimers']:
                config['simplified_interface'] = True
                config['tool_count'] = 4 # Only basic tools
                config['auto_save'] = True
            elif condition == 'parkinsons':
                config['motion_stabilization'] = True
                config['larger_buttons'] = True
            elif condition == 'stroke_recovery':
                config['one_handed_mode'] = True
                config['adaptive_difficulty'] = True

```

```

# Adjust for age
if user_profile.get('age', 30) > 65:
    config['button_size'] = 'large'
    config['contrast_ratio'] = 'high'
    config['font_size'] = 'large'

return config

def initialize_tools(self, user_profile):
    """
    Initialize drawing tools based on user capabilities
    """
    tools = {
        'basic': {
            'pencil': {
                'sizes': [1, 3, 5, 8],
                'pressure_sensitive': False,
                'stabilizer': True
            },
            'brush': {
                'types': ['round', 'flat', 'watercolor'],
                'sizes': [2, 5, 10, 20],
                'transparency': True
            },
            'eraser': {
                'sizes': [5, 10, 20, 40],
                'shape': 'round'
            },
            'fill': {
                'tolerance': 20,
                'contiguous_only': True
            }
        },
        'shapes': {
            'rectangle': {
                'fill': True,
                'stroke': True,
                'rounded_corners': True
            },
            'circle': {
                'fill': True,
                'stroke': True
            },
            'line': {
                'arrowheads': True,
                'dashed': True
            },
            'polygon': {
                'sides': 3,
                'regular': True
            }
        },
        'advanced': {
            'stamp': {
                'categories': ['animals', 'nature', 'emotions', 'objects'],
                'size_adjustable': True,
                'rotate': True
            },
            'text': {
                'fonts': self.select_accessible_fonts(user_profile),
                'sizes': [12, 16, 24, 32, 48],
                'colors': self.get_high_contrast_colors()
            }
        }
    }

```

```

        'gradient': {
            'linear': True,
            'radial': True,
            'color_picker': 'advanced'
        }
    }
}

# Simplify tools for certain users
if user_profile.get('age', 30) > 70:
    tools = self.simplify_tools_for_elderly(tools)

    if user_profile.get('cognitive_condition') in ['dementia',
'alzheimers']:
        tools = self.simplify_tools_for_dementia(tools)

return tools

def load_coloring_books(self, user_profile):
    """
    Load age-appropriate coloring books
    """
    coloring_books = {
        'elderly': {
            'themes': ['nature', 'animals', 'mandalas', 'memory_lane'],
            'complexity': 'low',
            'line_thickness': 'medium',
            'therapeutic': True,
            'examples': [
                {
                    'title': 'Garden Memories',
                    'description': 'Beautiful garden scenes for
reminiscence',

                    'pages': 12,
                    'preview_colors': ['pastel', 'vibrant']
                },
                {
                    'title': 'Animal Friends',
                    'description': 'Friendly animals with large coloring
areas',

                    'pages': 8,
                    'preview_colors': ['natural', 'fantasy']
                }
            ]
        },
        'children': {
            'themes': ['cartoons', 'fairytales', 'vehicles', 'animals'],
            'complexity': 'very_low',
            'line_thickness': 'thick',
            'educational': True,
            'examples': [
                {
                    'title': 'Magical Forest',
                    'description': 'Friendly forest creatures',
                    'pages': 10,
                    'interactive': True // Makes sounds when colored
                },
                {
                    'title': 'ABC Coloring',
                    'description': 'Learn letters while coloring',
                    'pages': 26,
                    'educational_value': 'alphabet_learning'
                }
            ]
        }
    }

```

```

    },
    'therapeutic': {
        'themes': ['stress_relief', 'mindfulness',
'emotional_expression'],
        'complexity': 'variable',
        'line_thickness': 'variable',
        'guided': True,
        'examples': [
            {
                'title': 'Emotion Garden',
                'description': 'Color based on your emotions',
                'pages': 6,
                'emotion_guided': True
            },
            {
                'title': 'Calm Patterns',
                'description': 'Repetitive patterns for mindfulness',
                'pages': 8,
                'breathing_guided': True
            }
        ]
    }
}

```

```

# Select appropriate coloring books
age = user_profile.get('age', 30)
condition = user_profile.get('cognitive_condition')

if age > 65 or condition in ['dementia', 'alzheimers']:
    return coloring_books['elderly']
elif age < 12:
    return coloring_books['children']
else:
    return coloring_books['therapeutic']

def setup_accessibility(self, user_profile):
    """
    Setup accessibility features based on user needs
    """
    features = {
        'voice_control': False,
        'eye_tracking': False,
        'switch_control': False,
        'screen_reader': False,
        'high_contrast': False,
        'motion_reduction': False
    }

    # Enable features based on needs
    if user_profile.get('motor_impairment'):
        features['voice_control'] = True
        features['switch_control'] = True

    if user_profile.get('visual_impairment'):
        features['high_contrast'] = True
        features['screen_reader'] = True

    if user_profile.get('cognitive_condition') in ['adhd', 'autism']:
        features['motion_reduction'] = True
        features['simplified_interface'] = True

    return features

```

B. Drawing AI Assistant with Emotional Intelligence

python

```
class DrawingAIAssistant:
    def __init__(self):
        self.style_recommender = StyleRecommender()
        self.color_adviser = ColorAdviser()
        self.creative_companion = CreativeCompanion()
        self.emotional_responder = DrawingEmotionalResponder()

    def provide_assistance(self, drawing_context, user_profile):
        """
        Provide appropriate drawing assistance based on context
        """
        assistance = {
            'suggestions': [],
            'encouragement': [],
            'technical_help': [],
            'creative_ideas': []
        }

        # Check if user is struggling
        if self.detect_struggle(drawing_context):
            assistance['suggestions'].append(
                self.suggest_simplified_approach(drawing_context)
            )
            assistance['encouragement'].append(
                self.generate_encouragement(drawing_context, user_profile)
            )

        # Check if user might want creative ideas
        if self.detect_creative_block(drawing_context):
            assistance['creative_ideas'].append(
                self.suggest_creative_direction(drawing_context, user_profile)
            )

        # Provide color suggestions
        assistance['color_suggestions'] = self.color_adviser.suggest_colors(
            drawing_context, user_profile
        )

        # Provide style suggestions
        assistance['style_suggestions'] = self.style_recommender.suggest_style(
            drawing_context, user_profile
        )

        return assistance

    def detect_struggle(self, drawing_context):
        """
        Detect if user is struggling with drawing
        """
        indicators = [
            'frequent_erasing',
            'long_pauses',
            'repetitive_strokes',
            'abandoned_drawings',
            'frustrated_gestures' # From camera if available
        ]

        return any(indicator in drawing_context for indicator in indicators)

    def suggest_simplified_approach(self, drawing_context):
```

```

"""
Suggest simplified drawing approach
"""
simplifications = {
    'complex_object': "Try breaking it down into basic shapes first",
    'perspective': "Don't worry about perfect perspective - focus on
what you enjoy",
    'details': "Sometimes less is more. Try a simpler version first",
    'coloring': "You could try filling larger areas with one color
first"
}

# Identify what user is trying to draw
attempted_subject = self.identify_drawing_subject(drawing_context)

if attempted_subject in simplifications:
    return simplifications[attempted_subject]
else:
    return "Try starting with the biggest, simplest shapes first"

def generate_encouragement(self, drawing_context, user_profile):
    """
    Generate personalized encouragement
    """
    encouragements = {
        'elderly': [
            "Your experience brings unique beauty to your art",
            "Every stroke tells a story - I love seeing yours",
            "Art is about expression, not perfection"
        ],
        'children': [
            "Wow, you're doing amazing!",
            "I love your creative choices!",
            "You're such a talented artist!"
        ],
        'therapeutic': [
            "Art is a wonderful way to express yourself",
            "There's no wrong way to create art",
            "I can see you putting your heart into this"
        ]
    }

    age = user_profile.get('age', 30)
    if age > 65:
        category = 'elderly'
    elif age < 13:
        category = 'children'
    else:
        category = 'therapeutic'

    return random.choice(encouragements[category])

def celebrate_completion(self, drawing, user_profile):
    """
    Celebrate when user completes a drawing
    """
    celebration = {
        'expression': 'excited',
        'visual_effects': ['confetti', 'sparkles'],
        'message': self.generate_completion_message(drawing, user_profile),
        'suggest_next': self.suggest_next_activity(drawing, user_profile)
    }

    # Trigger Marvin's happy expression

```



```

self.trigger_marvin_expression('happy', intensity=0.9)

# Show fireworks
self.trigger_visual_effect('fireworks')

return celebration

```

3. Virtual Tea Party System

A. Interactive Tea Party Environment

python

```

class VirtualTeaParty:
    def __init__(self, user_profile):
        self.user_profile = user_profile
        self.environment = self.setup_environment(user_profile)
        self.participants = self.initialize_participants(user_profile)
        self.conversation_engine = TeaPartyConversationEngine()
        self.activities = self.load_activities(user_profile)
        self.mood_atmosphere = self.set_mood_atmosphere(user_profile)

    def setup_environment(self, user_profile):
        """
        Create tea party environment based on user preferences
        """
        environments = {
            'traditional_english': {
                'table': 'wooden_round',
                'tea_set': 'porcelain_floral',
                'background': 'english_garden',
                'music': 'classical_soft',
                'lighting': 'afternoon_sun'
            },
            'japanese_tea_ceremony': {
                'table': 'low_tatami',
                'tea_set': 'japanese_ceremonial',
                'background': 'zen_garden',
                'music': 'shakuhachi',
                'lighting': 'soft_tatami'
            },
            'childrens_party': {
                'table': 'colorful_plastic',
                'tea_set': 'cartoon_characters',
                'background': 'rainbow_playroom',
                'music': 'happy_playful',
                'lighting': 'bright_cheerful'
            },
            'memory_lane': {
                'table': 'vintage_wood',
                'tea_set': 'grandmas_china',
                'background': 'old_photographs',
                'music': 'era_appropriate',
                'lighting': 'warm_nostalgic'
            }
        }

        # Select environment based on age and preferences
        age = user_profile.get('age', 30)
        if age > 70:
            return environments['memory_lane']

```

```

elif age < 12:
    return environments['childrens_party']
else:
    return environments['traditional_english']

def initialize_participants(self, user_profile):
    """
    Create tea party participants (AI characters)
    """
    participants = []

    # Base participant for elderly
    if user_profile.get('age', 30) > 70:
        participants.append({
            'name': 'Eleanor',
            'age': 'similar',
            'personality': 'gentle_listener',
            'conversation_style': 'reminiscent',
            'appearance': 'kind_elderly_woman',
            'role': 'companion'
        })

    # Add memory-triggering characters
    if user_profile.get('memory_triggers'):
        for trigger in user_profile['memory_triggers']:
            participants.append({
                'name': trigger['name'],
                'persona': trigger['persona'],
                'purpose': 'memory_evocation'
            })

    # Base participant for children
    elif user_profile.get('age', 30) < 12:
        participants.append({
            'name': 'Mr. Teapot',
            'age': 'ageless',
            'personality': 'funny_friendly',
            'conversation_style': 'playful_educational',
            'appearance': 'animated_teapot',
            'role': 'fun_companion'
        })

    participants.append({
        'name': 'Sugar Cube',
        'age': 'young',
        'personality': 'sweet_energetic',
        'conversation_style': 'simple_playful',
        'appearance': 'cute_sugar_cube',
        'role': 'play_friend'
    })

    # Add AI companion (Marvin)
    participants.append({
        'name': 'Marvin',
        'personality': 'caring_companion',
        'conversation_style': 'adaptive',
        'appearance': 'friendly_robot',
        'role': 'host_companion'
    })

    return participants

def load_activities(self, user_profile):
    """

```

```

Load age-appropriate tea party activities
"""
activities = {
    'pour_tea': {
        'difficulty': 'simple',
        'motor_skills': 'basic',
        'description': 'Pour virtual tea into cups',
        'therapeutic_value': 'fine_motor_practice'
    },
    'serve_cake': {
        'difficulty': 'simple',
        'motor_skills': 'basic',
        'description': 'Serve virtual cakes to participants',
        'social_value': 'sharing_practice'
    },
    'conversation_game': {
        'difficulty': 'variable',
        'cognitive_skills': 'conversational',
        'description': 'Engage in guided conversation',
        'social_value': 'communication_practice'
    },
    'memory_sharing': {
        'difficulty': 'variable',
        'cognitive_skills': 'recall',
        'description': 'Share memories prompted by objects',
        'therapeutic_value': 'reminiscence_therapy'
    },
    'tea_blending': {
        'difficulty': 'medium',
        'cognitive_skills': 'creativity',
        'description': 'Create custom tea blends',
        'educational_value': 'sensory_exploration'
    }
}

# Filter activities based on user capabilities
filtered_activities = {}
for name, activity in activities.items():
    if self.is_activity_appropriate(activity, user_profile):
        filtered_activities[name] = activity

return filtered_activities

def is_activity_appropriate(self, activity, user_profile):
    """
    Check if activity is appropriate for user
    """
    # Check cognitive requirements
    if 'cognitive_skills' in activity:
        required_skill = activity['cognitive_skills']
        user_cognitive = user_profile.get('cognitive_level', 'normal')

        if required_skill == 'recall' and
user_profile.get('memory_impairment'):
            return False

    # Check motor requirements
    if 'motor_skills' in activity:
        required_motor = activity['motor_skills']
        user_motor = user_profile.get('motor_ability', 'normal')

        if required_motor == 'fine' and
user_profile.get('motor_impairment'):
            return False

```

```
return True
```

B. Tea Party Conversation Engine

python

```
class TeaPartyConversationEngine:
    def __init__(self):
        self.conversation_topics = self.load_topics()
        self.prompt_generator = ConversationPromptGenerator()
        self.emotional_regulator = ConversationEmotionalRegulator()
        self.memory_trigger = MemoryTriggerEngine()

    def load_topics(self):
        """
        Load age-appropriate conversation topics
        """
        return {
            'elderly': {
                'reminiscence': [
                    "Tell me about your favorite childhood memory",
                    "What was the first concert you ever attended?",
                    "Describe your first job",
                    "What was school like when you were young?"
                ],
                'current_interests': [
                    "What do you enjoy doing these days?",
                    "Tell me about your family",
                    "What's your favorite season and why?"
                ],
                'wisdom_sharing': [
                    "What's the best advice you've ever received?",
                    "What life lesson would you share with younger people?"
                ]
            },
            'children': {
                'imagination': [
                    "If you could have any superpower, what would it be?",
                    "What would your perfect day look like?",
                    "If you were an animal, what would you be?"
                ],
                'playful': [
                    "What's the funniest thing that's happened to you?",
                    "What's your favorite silly joke?",
                    "If your toys could talk, what would they say?"
                ],
                'learning': [
                    "What's something new you learned recently?",
                    "What's your favorite thing about school?",
                    "What do you want to be when you grow up?"
                ]
            },
            'intergenerational': {
                'shared_experiences': [
                    "What's your favorite thing about each season?",
                    "What makes a good friend?",
                    "What's the best meal you've ever had?"
                ],
                'comparisons': [
                    "How is life different now than when you were young?",
                    "What games did you play as a child?"
                ]
            }
        }
```

```

    }
}

def generate_conversation_flow(self, user_profile, participants):
    """
    Generate a natural conversation flow for the tea party
    """
    # Select topic category based on user
    age = user_profile.get('age', 30)
    if age > 65:
        category = 'elderly'
    elif age < 13:
        category = 'children'
    else:
        category = 'intergenerational'

    # Create conversation structure
    conversation = {
        'opening': self.generate_opening(category),
        'main_topics': self.select_topics(category, 3),
        'questions_for_user': self.generate_questions(category,
user_profile),
        'ai_responses': self.prepare_ai_responses(category),
        'closing': self.generate_closing(category)
    }

    # Add memory triggers for elderly
    if age > 70 and user_profile.get('memory_triggers'):
        conversation['memory_prompts'] = self.generate_memory_prompts(
            user_profile['memory_triggers']
        )

    return conversation

def generate_memory_prompts(self, memory_triggers):
    """
    Generate prompts to trigger positive memories
    """
    prompts = []

    for trigger in memory_triggers:
        if trigger['type'] == 'object':
            prompts.append({
                'trigger_object': trigger['object'],
                'prompt': f"This {trigger['object']} reminds me of...",
                'visual_cue': trigger.get('image'),
                'sensory_cue': trigger.get('sensory_description')
            })
        elif trigger['type'] == 'topic':
            prompts.append({
                'topic': trigger['topic'],
                'prompt': f"Tell me about your experiences with
{trigger['topic']}",
                'guiding_questions': trigger.get('questions', [])
            })

    return prompts

```

4. Integration & Web Interface

A. Unified Web Portal Interface

html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Marvin's Interactive Playroom</title>
  <link rel="stylesheet" href="styles/playroom.css">
</head>
<body>
  <div class="playroom-container">
    <!-- Marvin's Expression Display -->
    <div class="marvin-expression-display">
      <canvas id="marvinFace" width="800" height="600"></canvas>
      <div class="expression-controls">
        <button onclick="marvin.setExpression('happy')">? Happy</button>
        <button onclick="marvin.setExpression('excited')">?
Excited</button>
        <button onclick="marvin.setExpression('thinking')">?
Thinking</button>
        <button onclick="marvin.robotDance()">? Robot Dance!</button>
        <button onclick="marvin.fireworks()">? Fireworks!</button>
      </div>
    </div>

    <!-- Activity Selection -->
    <div class="activity-grid">
      <!-- Drawing Program -->
      <div class="activity-card" onclick="startDrawingProgram()">
        <div class="activity-icon">?</div>
        <h3>Creative Drawing</h3>
        <p>Draw, color, and create with AI assistance</p>
        <div class="activity-features">
          <span>Coloring Books</span>
          <span>AI Assistant</span>
          <span>Accessible Tools</span>
        </div>
      </div>

      <!-- Tea Party -->
      <div class="activity-card" onclick="startTeaParty()">
        <div class="activity-icon">☕</div>
        <h3>Virtual Tea Party</h3>
        <p>Join Marvin for tea and conversation</p>
        <div class="activity-features">
          <span>AI Companions</span>
          <span>Memory Sharing</span>
          <span>Social Games</span>
        </div>
      </div>

      <!-- Memory Games -->
      <div class="activity-card" onclick="startMemoryGames()">
        <div class="activity-icon">?</div>
        <h3>Memory Games</h3>
        <p>Brain exercises for all ages</p>
        <div class="activity-features">
          <span>Adaptive Difficulty</span>

```

```

        <span>Progress Tracking</span>
        <span>Therapeutic</span>
    </div>
</div>

<!-- Social Games -->
<div class="activity-card" onclick="startSocialGames()">
    <div class="activity-icon">?</div>
    <h3>Social Games</h3>
    <p>Play with AI or friends</p>
    <div class="activity-features">
        <span>2-Player Games</span>
        <span>AI Opponent</span>
        <span>Cooperative Play</span>
    </div>
</div>
</div>

<!-- Current Session Display -->
<div class="current-session">
    <h2>Your Current Activity</h2>
    <div id="activityContainer">
        <!-- Activity will load here -->
    </div>
</div>

<!-- Marvin's Status Panel -->
<div class="marvin-status">
    <div class="status-indicators">
        <div class="status-item">
            <span class="status-label">Mood:</span>
            <span class="status-value" id="marvinMood">Happy</span>
        </div>
        <div class="status-item">
            <span class="status-label">Energy:</span>
            <span class="status-value" id="marvinEnergy">High</span>
        </div>
        <div class="status-item">
            <span class="status-label">Connection:</span>
            <span class="status-value"
id="marvinConnection">Strong</span>
        </div>
    </div>
    <div class="marvin-message" id="marvinMessage">
        Ready to play! Choose an activity or just chat with me!
    </div>
</div>
</div>

<!-- WebSocket Connection -->
<script src="js/websocket.js"></script>
<!-- Marvin Expression Engine -->
<script src="js/expression-engine.js"></script>
<!-- Activity Loaders -->
<script src="js/activity-loader.js"></script>

<script>
    // Initialize Marvin's expression display
    const marvin = new WebExpressionDisplay('marvinFace');

    // Connect to WebSocket for real-time updates
    const ws = new ExpressionWebSocket('ws://localhost:8765');

    // Start activities

```

```

function startDrawingProgram() {
    loadActivity('drawing', {
        userProfile: getUserProfile(),
        accessibility: getAccessibilitySettings()
    });
}

function startTeaParty() {
    loadActivity('tea-party', {
        userProfile: getUserProfile(),
        participants: selectTeaPartyGuests(),
        theme: selectTeaPartyTheme()
    });
}

// Mirror user's detected emotions
function mirrorUserEmotion(emotion) {
    ws.sendUserEmotion(emotion);
}

// Update Marvin's status based on activity
function updateMarvinStatus(activity, userEngagement) {
    const status = calculateMarvinStatus(activity, userEngagement);

    document.getElementById('marvinMood').textContent = status.mood;
    document.getElementById('marvinEnergy').textContent = status.energy;
    document.getElementById('marvinMessage').textContent =
status.message;

    // Update expression
    marvin.setExpression(status.expression, status.intensity);
}
</script>
</body>
</html>

```

5. Implementation Strategy

Phase 1: Foundation (Weeks 1-2)

text

Objectives:

- Create basic expression engine with core emotions
- Implement simple drawing canvas
- Setup WebSocket communication
- Create basic web interface

Deliverables:

- Marvin's face with 5 basic expressions
- Simple drawing tools (pencil, eraser, fill)
- Real-time WebSocket connection
- Basic activity selection page

Phase 2: Advanced Expressions & Drawing (Weeks 3-4)

text

Objectives:

- Add complex expressions with animations

- Implement coloring book system
- Add accessibility features
- Create tea party environment

Deliverables:

- Animated expressions with accessories
- Multiple coloring book themes
- Voice control for drawing
- Basic tea party with one AI companion

Phase 3: Emotional Intelligence & Social Features (Weeks 5-6)

text

Objectives:

- Implement mirror neuron system
- Add emotional scaffolding
- Create conversation engine
- Implement multiplayer features

Deliverables:

- Emotion mirroring based on user state
- Therapeutic conversation flows
- 2-player drawing collaboration
- Tea party with multiple AI characters

Phase 4: Polish & Integration (Weeks 7-8)

text

Objectives:

- Optimize for RPi5 performance
- Add visual effects (fireworks, sparkles)
- Integrate with existing Marvin system
- User testing and refinement

Deliverables:

- Smooth 60fps animations on RPi5
- Celebration effects system
- Shared memory with main Marvin
- User-tested refined interface

6. Technical Requirements & Optimization

A. RPi5 Performance Optimization

python

```
class RPIOptimization:
    def __init__(self):
        self.performance_profiles = {
            'normal': {
                'frame_rate': 30,
                'resolution': '800x600',
                'particle_count': 50,
                'animation_complexity': 'medium'
            },
            'reduced': {
                'frame_rate': 15,
```

```

        'resolution': '640x480',
        'particle_count': 20,
        'animation_complexity': 'low'
    },
    'minimal': {
        'frame_rate': 10,
        'resolution': '480x320',
        'particle_count': 5,
        'animation_complexity': 'basic'
    }
}

def monitor_performance(self):
    """
    Monitor system performance and adjust graphics accordingly
    """
    cpu_temp = self.get_cpu_temp()
    cpu_usage = self.get_cpu_usage()
    memory_available = self.get_available_memory()

    if cpu_temp > 70 or cpu_usage > 85:
        return 'minimal'
    elif cpu_temp > 60 or cpu_usage > 70:
        return 'reduced'
    else:
        return 'normal'

def optimize_animations(self, performance_profile):
    """
    Optimize animations for current performance profile
    """
    profile = self.performance_profiles[performance_profile]

    # Adjust animation properties
    self.animation_engine.set_frame_rate(profile['frame_rate'])
    self.canvas.set_resolution(profile['resolution'])
    self.particle_system.set_max_particles(profile['particle_count'])

    # Simplify complex animations
    if profile['animation_complexity'] == 'basic':
        self.disable_complex_effects()
        self.simplify_expressions()

```

B. Accessibility Configuration

json

```

{
  "accessibility_profiles": {
    "elderly": {
      "interface_size": "large",
      "contrast": "high",
      "animation_speed": "slow",
      "sound_volume": "higher",
      "tooltip_duration": "long",
      "confirmation_prompts": true
    },
    "dementia": {
      "interface_size": "extra_large",
      "contrast": "maximum",
      "animation_speed": "very_slow",
      "repetition_allowed": true,
      "guided_mode": true,

```

```

    "distraction_free": true
  },
  "children": {
    "interface_size": "medium",
    "contrast": "vibrant",
    "animation_speed": "medium",
    "sound_effects": true,
    "reward_animations": true,
    "guided_instructions": true
  },
  "motor_impairment": {
    "interface_size": "custom",
    "click_assist": true,
    "voice_control": true,
    "gesture_control": false,
    "switch_access": true,
    "dwell_control": true
  }
}
}

```

7. Therapeutic Benefits

A. For Elderly with Dementia:

1. **Familiar Activities:** Tea party triggers positive memories
2. **Cognitive Stimulation:** Simple games maintain neural pathways
3. **Emotional Connection:** Marvin's expressions provide companionship
4. **Fine Motor Maintenance:** Drawing activities maintain dexterity

B. For Children:

1. **Social Skills Development:** Tea party teaches conversation
2. **Emotional Intelligence:** Marvin models emotional responses
3. **Creativity Enhancement:** Drawing with AI encouragement
4. **Play-Based Learning:** Educational elements in fun activities

C. For Neurodivergent Individuals:

1. **Predictable Social Interactions:** AI provides consistent responses
2. **Emotional Regulation:** Marvin helps identify and manage emotions
3. **Structured Play:** Clear rules and expectations
4. **Safe Social Practice:** Low-pressure social interaction

Conclusion

This **Multi-Modal Interactive Engagement System** creates a comprehensive therapeutic and recreational environment that extends Marvin's capabilities far beyond conversation. By integrating:

1. **Emotional Expression Engine:** Visual feedback through animated expressions
2. **Mirror Neuron System:** Emotional synchronization for connection

3. **Creative Tools:** Accessible drawing and coloring for all abilities
4. **Social Activities:** Tea parties and games for engagement
5. **Therapeutic Design:** Age-appropriate and condition-specific adaptations

We transform the RPi5 into a **warm, expressive companion** that can:

- **Mirror emotions** to create connection
- **Guide therapeutic activities** for cognitive maintenance
- **Provide joyful interactions** through games and creativity
- **Adapt in real-time** to user's emotional and cognitive state

The system maintains the core **Peter Protocol ethics** of protection and dignity while adding layers of **emotional intelligence** and **therapeutic value**. Marvin becomes not just a guardian, but a **playful companion, creative partner, and empathetic friend**.

Key Innovation: The **Expression Engine** with **Mirror Neuron Integration** creates a uniquely human-like connection that helps users, especially those with cognitive or social challenges, understand and regulate their own emotions through Marvin's expressive visual feedback.

Augmented Reality Tea Party with Doll/Teddy Integration & Emotion Mirroring

1. Computer Vision System for Toy Recognition & Interaction

A. Toy Recognition & Pose Estimation System

python

```
class ToyRecognitionEngine:
    def __init__(self):
        self.toy_database = ToyDatabase()
        self.pose_estimator = ToyPoseEstimator()
        self.facial_recognition = ToyFacialRecognition()
        self.movement_tracker = MovementTracker()

    def initialize_childs_toys(self, camera_feed):
        """
        Scan the environment and identify all toys placed for the tea party
        """
        # Step 1: Detect all toys in scene
        toys = self.detect_all_toys(camera_feed)

        # Step 2: Identify each toy type and assign personality
        for toy in toys:
            toy['identity'] = self.identify_toy(toy['image'])
            toy['personality'] = self.assign_personality(toy['identity'])
            toy['role'] = self.assign_tea_party_role(toy['identity'])
            toy['voice'] = self.generate_toy_voice(toy['personality'])
```

```

        return toys

def detect_all_toys(self, frame):
    """
    Use computer vision to detect toys in the scene
    """
    # Using YOLO or similar for object detection
    detections = self.object_detector.detect(
        frame,
        classes=['teddy_bear', 'doll', 'stuffed_animal', 'action_figure']
    )

    toys = []
    for detection in detections:
        toy = {
            'bbox': detection['bbox'],
            'type': detection['class'],
            'confidence': detection['confidence'],
            'center': self.calculate_center(detection['bbox']),
            'rotation': self.estimate_rotation(detection['bbox'], frame)
        }

        # Crop toy image for detailed analysis
        toy['image'] = self.crop_image(frame, detection['bbox'])

        # Track toy movement over time
        toy['movement_history'] =
self.movement_tracker.initialize_track(toy['center'])

        toys.append(toy)

    return toys

def assign_personality(self, toy_identity):
    """
    Assign personality to each toy based on appearance
    """
    personalities = {
        'teddy_bear': {
            'gentle_brown': {
                'name': 'Mr. Snuggles',
                'personality': 'wise_elderly',
                'voice_pitch': 'low_warm',
                'speaking_style': 'slow_thoughtful',
                'favorite_topics': ['hugs', 'bedtime stories', 'honey']
            },
            'colorful_patchwork': {
                'name': 'Patches',
                'personality': 'cheerful_energetic',
                'voice_pitch': 'medium_chipper',
                'speaking_style': 'fast_excited',
                'favorite_topics': ['adventures', 'colors', 'games']
            }
        },
        'doll': {
            'princess_doll': {
                'name': 'Princess Sparkle',
                'personality': 'elegant_gracious',
                'voice_pitch': 'high_clear',
                'speaking_style': 'polished_refined',
                'favorite_topics': ['royal balls', 'magic', 'kindness']
            },
            'baby_doll': {

```

```

        'name': 'Baby Giggles',
        'personality': 'playful_infant',
        'voice_pitch': 'very_high_cute',
        'speaking_style': 'simple_repetitive',
        'favorite_topics': ['games', 'funny sounds', 'playtime']
    }
}

# Determine which specific type within category
specific_type = self.classify_toy_subtype(toy_identity)
return personalities.get(toy_identity['type'], {}).get(
    specific_type,
    self.default_personality(toy_identity['type'])
)

def generate_toy_voice(self, personality):
    """
    Generate unique voice characteristics for each toy
    """
    voice_templates = {
        'wise_elderly': {
            'pitch_variation': 0.1,
            'speech_rate': 0.7,
            'breathiness': 0.3,
            'warmth': 0.9,
            'unique_features': ['gentle_laugh', 'thoughtful_pauses']
        },
        'cheerful_energetic': {
            'pitch_variation': 0.4,
            'speech_rate': 1.2,
            'breathiness': 0.1,
            'warmth': 0.8,
            'unique_features': ['giggles', 'excited_gasps']
        },
        'elegant_gracious': {
            'pitch_variation': 0.2,
            'speech_rate': 1.0,
            'breathiness': 0.1,
            'warmth': 0.7,
            'unique_features': ['polite_laughter', 'precise_articulation']
        }
    }

    return voice_templates.get(
        personality['personality'],
        voice_templates['cheerful_energetic']
    )

```

B. Real-Time Toy Interaction Tracking

python

```

class ToyInteractionTracker:
    def __init__(self):
        self.interaction_history = []
        self.toy_states = {}
        self.child_toy_interactions = []

    def track_child_toy_interactions(self, child_pose, toy_positions):
        """
        Track how child interacts with each toy
        """

```

```

interactions = []

for toy_id, toy_position in toy_positions.items():
    # Calculate distance between child's hands and toy
    for hand in ['left_hand', 'right_hand']:
        hand_position = child_pose.get(hand)
        if hand_position:
            distance = self.calculate_distance(hand_position,
toy_position)

            if distance < 50: # pixels - toy is being touched
                interaction = {
                    'toy_id': toy_id,
                    'hand': hand,
                    'interaction_type': self.determine_interaction_type(
                        child_pose, toy_position
                    ),
                    'duration':
self.measure_interaction_duration(toy_id, hand),
                    'intensity':
self.calculate_interaction_intensity(child_pose)
                }

                interactions.append(interaction)

                # Update toy's emotional state based on interaction
                self.update_toy_emotional_state(toy_id, interaction)

    return interactions

def determine_interaction_type(self, child_pose, toy_position):
    """
    Determine type of interaction (hugging, moving, feeding, etc.)
    """
    # Analyze hand shape and movement
    hand_shape = self.analyze_hand_shape(child_pose)
    movement_pattern = self.analyze_movement_pattern(child_pose)

    if hand_shape == 'open_palm' and movement_pattern == 'patting':
        return 'petting'
    elif hand_shape == 'grasp' and movement_pattern == 'lifting':
        return 'picking_up'
    elif hand_shape == 'cup' and movement_pattern == 'moving_toward_mouth':
        return 'feeding'
    elif hand_shape == 'wrap' and movement_pattern == 'squeeze':
        return 'hugging'
    elif hand_shape == 'point' and movement_pattern == 'still':
        return 'pointing_at'

    return 'touching'

def update_toy_emotional_state(self, toy_id, interaction):
    """
    Update toy's emotional state based on how child interacts with it
    """
    if toy_id not in self.toy_states:
        self.toy_states[toy_id] = {
            'happiness': 0.5,
            'energy': 0.5,
            'engagement': 0.5,
            'preferred_interactions': []
        }

    toy_state = self.toy_states[toy_id]

```

```

# Adjust emotional state based on interaction
if interaction['interaction_type'] in ['hugging', 'petting']:
    toy_state['happiness'] = min(1.0, toy_state['happiness'] + 0.1)
    toy_state['engagement'] = min(1.0, toy_state['engagement'] + 0.15)
elif interaction['interaction_type'] == 'feeding':
    toy_state['energy'] = min(1.0, toy_state['energy'] + 0.2)

# Record preferred interactions
if interaction['duration'] > 2.0: # Longer interactions indicate
preference
    if interaction['interaction_type'] not in
toy_state['preferred_interactions']:

toy_state['preferred_interactions'].append(interaction['interaction_type'])

def generate_toy_response(self, toy_id, interaction):
    """
    Generate how toy would respond to the interaction
    """
    toy_state = self.toy_states[toy_id]
    personality = self.get_toy_personality(toy_id)

    responses = {
        'hugging': {
            'high_happiness': [
                "Oh, I love your hugs!",
                "You give the best hugs!",
                "Mmm, so cozy!"
            ],
            'medium_happiness': [
                "That's a nice hug!",
                "Thank you for the hug!",
                "Hugs make everything better!"
            ]
        },
        'petting': {
            'wise_elderly': [
                "A gentle pat always makes me feel cared for.",
                "Your kind touch reminds me of sunny days."
            ],
            'cheerful_energetic': [
                "Yay! Petting time!",
                "This feels wonderful! Do it again!"
            ]
        },
        'feeding': [
            "Yummy! Thank you for the tea!",
            "This is delicious! You're a great host!",
            "Mmm, my favorite!"
        ]
    }

# Select appropriate response
happiness_level = 'high_happiness' if toy_state['happiness'] > 0.8 else
'medium_happiness'

if interaction['interaction_type'] in responses:
    category_responses = responses[interaction['interaction_type']]

    if isinstance(category_responses, dict):
        # Has subcategories
        if happiness_level in category_responses:
            response_list = category_responses[happiness_level]

```



```

        elif personality['personality'] in category_responses:
            response_list =
category_responses[personality['personality']]
        else:
            response_list = list(category_responses.values())[0]
    else:
        response_list = category_responses

    return random.choice(response_list)

return "Thank you for playing with me!"

```

2. Advanced Facial Expression & Body Language Recognition

A. Multi-Modal Emotion Recognition

python

```

class ChildEmotionRecognizer:
    def __init__(self):
        self.facial_recognizer = FacialExpressionAnalyzer()
        self.body_language_analyzer = BodyLanguageAnalyzer()
        self.vocal_analyzer = VocalToneAnalyzer()
        self.context_analyzer = ContextAnalyzer()

    def analyze_child_state(self, camera_feed, audio_feed, context):
        """
        Comprehensive analysis of child's emotional state
        """
        # Multi-modal emotion recognition
        facial_emotion = self.facial_recognizer.analyze(camera_feed)
        body_emotion = self.body_language_analyzer.analyze(camera_feed)
        vocal_emotion = self.vocal_analyzer.analyze(audio_feed) if audio_feed
else None

        # Combine results with confidence weights
        emotions = []
        confidences = []

        if facial_emotion:
            emotions.append(facial_emotion['emotion'])
            confidences.append(facial_emotion['confidence'] * 0.4)

        if body_emotion:
            emotions.append(body_emotion['emotion'])
            confidences.append(body_emotion['confidence'] * 0.3)

        if vocal_emotion:
            emotions.append(vocal_emotion['emotion'])
            confidences.append(vocal_emotion['confidence'] * 0.3)

        # Determine primary emotion
        if emotions:
            primary_emotion = max(set(emotions), key=emotions.count)
            confidence = sum(confidences) / len(confidences)
        else:
            primary_emotion = 'neutral'
            confidence = 0.5

        # Adjust based on context
        context_adjusted = self.context_analyzer.adjust_emotion(

```

```

        primary_emotion,
        context,
        confidence
    )

    return {
        'primary_emotion': context_adjusted['emotion'],
        'confidence': context_adjusted['confidence'],
        'components': {
            'facial': facial_emotion,
            'body': body_emotion,
            'vocal': vocal_emotion
        },
        'intensity': self.calculate_emotion_intensity(
            facial_emotion, body_emotion, vocal_emotion
        ),
        'engagement_level': self.calculate_engagement_level(body_emotion),
        'micro_expressions': self.detect_micro_expressions(camera_feed)
    }

def analyze_body_language(self, pose_data):
    """
    Analyze body language for emotional cues
    """
    body_indicators = {
        'posture': {
            'leaning_forward': 'engaged_interested',
            'leaning_back': 'relaxed_contemplative',
            'slumped': 'tired_disengaged',
            'erect_tense': 'alert_anxious'
        },
        'arm_position': {
            'crossed': 'defensive_closed',
            'open': 'receptive_open',
            'behind_back': 'formal_reserved',
            'gesturing': 'expressive_engaged'
        },
        'head_tilt': {
            'right': 'curious_playful',
            'left': 'contemplative',
            'down': 'shy_submissive',
            'up': 'confident_proud'
        },
        'fidgeting': {
            'low': 'calm_focused',
            'medium': 'energetic_excited',
            'high': 'anxious_bored'
        }
    }

    # Calculate body language score
    body_emotion = {
        'engagement': 0.5,
        'openness': 0.5,
        'energy': 0.5,
        'confidence': 0.5
    }

    # Analyze each body part
    if pose_data.get('shoulders', {}).get('slumped', False):
        body_emotion['energy'] -= 0.3
        body_emotion['confidence'] -= 0.2

    if pose_data.get('arms', {}).get('crossed', False):

```

```

        body_emotion['openness'] -= 0.4

    if pose_data.get('head', {}).get('tilted', 'none') == 'right':
        body_emotion['engagement'] += 0.3

    # Calculate fidgeting
    fidget_score = self.calculate_fidgeting(pose_data)
    if fidget_score > 0.7:
        body_emotion['energy'] += 0.4
    elif fidget_score < 0.3:
        body_emotion['calmness'] = 0.8

    return body_emotion

def detect_micro_expressions(self, facial_data):
    """
    Detect brief, involuntary facial expressions
    """
    micro_expressions = []

    # Check for quick expression changes
    expression_history = self.get_recent_expressions(facial_data,
window_ms=500)

    for i in range(1, len(expression_history)):
        prev = expression_history[i-1]
        curr = expression_history[i]

        # If expression changed rapidly (micro-expression)
        if curr['timestamp'] - prev['timestamp'] < 200: # 200ms
            if curr['expression'] != prev['expression']:
                micro_expressions.append({
                    'from': prev['expression'],
                    'to': curr['expression'],
                    'duration': curr['timestamp'] - prev['timestamp'],
                    'intensity': curr['intensity']
                })

    return micro_expressions

```

B. Real-Time Emotion Mirroring with Delay & Amplification

python

```

class EmotionMirroringSystem:
    def __init__(self):
        self.mirroring_strategies = {
            'direct_mirror': {
                'delay_ms': 300,
                'amplification': 1.0,
                'use_for': ['happy', 'excited', 'surprised']
            },
            'complementary': {
                'delay_ms': 500,
                'amplification': 0.8,
                'use_for': ['sad', 'frustrated'],
                'target_emotion': 'compassionate'
            },
            'scaffolding': {
                'delay_ms': 1000,
                'amplification': 1.2,
                'use_for': ['angry', 'anxious'],
                'target_emotion': 'calm'
            }
        }

```

```

        },
        'exaggerated_playful': {
            'delay_ms': 200,
            'amplification': 1.5,
            'use_for': ['playful', 'silly'],
            'target_emotion': 'excited_playful'
        }
    }

    self.response_history = []

def generate_mirrored_response(self, child_emotion, child_profile):
    """
    Generate Marvin's emotional response to child's emotion
    """
    # Select mirroring strategy
    strategy = self.select_mirroring_strategy(child_emotion, child_profile)

    # Calculate response emotion
    if strategy['type'] == 'direct_mirror':
        response_emotion = child_emotion['primary_emotion']
        intensity = child_emotion['intensity'] * strategy['amplification']

    elif strategy['type'] == 'complementary':
        response_emotion = strategy['target_emotion']
        # Complementary emotions should match intensity but different
valence        intensity = child_emotion['intensity'] * 0.7

    elif strategy['type'] == 'scaffolding':
        response_emotion = strategy['target_emotion']
        # Scaffolding starts slightly lower intensity
        intensity = child_emotion['intensity'] * 0.5

    elif strategy['type'] == 'exaggerated_playful':
        response_emotion = strategy['target_emotion']
        intensity = min(1.0, child_emotion['intensity'] * 1.5)

    # Add delay for natural response
    delay = random.gauss(strategy['delay_ms'], strategy['delay_ms'] * 0.2)

    # Generate facial expression sequence
    expression_sequence = self.generate_expression_sequence(
        response_emotion,
        intensity,
        delay
    )

    # Add conversational response
    verbal_response = self.generate_verbal_mirroring(
        child_emotion,
        response_emotion,
        child_profile
    )

    return {
        'expression_sequence': expression_sequence,
        'verbal_response': verbal_response,
        'delay_ms': delay,
        'strategy_used': strategy['type'],
        'intensity_match': child_emotion['intensity'] / intensity if
intensity > 0 else 0
    }

```

```

def generate_expression_sequence(self, emotion, intensity, delay_ms):
    """
    Generate natural-looking sequence of facial expressions
    """
    sequences = {
        'happy': [
            {'expression': 'surprised', 'duration': 100, 'intensity':
intensity * 0.3},
            {'expression': 'happy', 'duration': 300, 'intensity': intensity
* 0.8},
            {'expression': 'happy', 'duration': 2000, 'intensity':
intensity},
            {'expression': 'content', 'duration': 1500, 'intensity':
intensity * 0.6}
        ],
        'excited': [
            {'expression': 'surprised', 'duration': 50, 'intensity':
intensity * 0.5},
            {'expression': 'excited', 'duration': 400, 'intensity':
intensity},
            {'expression': 'excited', 'duration': 1000, 'intensity':
intensity * 0.9},
            {'expression': 'happy', 'duration': 2000, 'intensity': intensity
* 0.7}
        ],
        'compassionate': [
            {'expression': 'concerned', 'duration': 300, 'intensity':
intensity * 0.6},
            {'expression': 'gentle_smile', 'duration': 2000, 'intensity':
intensity},
            {'expression': 'warm', 'duration': 1500, 'intensity': intensity
* 0.8}
        ],
        'playful_surprised': [
            {'expression': 'surprised', 'duration': 150, 'intensity':
intensity},
            {'expression': 'thinking', 'duration': 200, 'intensity':
intensity * 0.5},
            {'expression': 'playful', 'duration': 800, 'intensity':
intensity * 0.9},
            {'expression': 'happy', 'duration': 1000, 'intensity': intensity
* 0.7}
        ]
    }

    sequence = sequences.get(emotion, sequences['happy'])

    # Add initial delay
    sequence.insert(0, {
        'expression': 'neutral',
        'duration': delay_ms,
        'intensity': 0.1
    })

    return sequence

def generate_verbal_mirroring(self, child_emotion, response_emotion,
child_profile):
    """
    Generate verbal responses that mirror child's emotion
    """
    verbal_mirrors = {
        'happy': {
            'children': [

```

```

        "You look so happy! That makes me happy too!",
        "I love seeing you smile!",
        "Your happiness is contagious!"
    ],
    'elderly': [
        "That beautiful smile brightens the whole room.",
        "Seeing you happy warms my heart.",
        "Your joy is a gift to everyone around you."
    ]
},
'excited': {
    'children': [
        "Wow, you're so excited! What's the exciting news?",
        "I can feel your excitement! Tell me all about it!",
        "Your excitement is getting me excited too!"
    ],
    'elderly': [
        "I can see the sparkle in your eyes!",
        "Your enthusiasm is wonderful to see.",
        "That excitement is bringing energy to our tea party!"
    ]
},
'sad': {
    'children': [
        "I see you're feeling sad. I'm here with you.",
        "It's okay to feel sad sometimes. Want to talk about it?",
        "I'm sending you a virtual hug."
    ],
    'elderly': [
        "I can see something's weighing on your heart. I'm
listening.",
        "Sometimes a cup of tea and a friend can help.",
        "You're not alone in this feeling."
    ]
},
'playful': {
    'children': [
        "You're being so playful! I love it!",
        "What mischief are you planning?",
        "Your playfulness is infectious!"
    ],
    'elderly': [
        "That playful spirit keeps you young at heart!",
        "I love seeing your playful side!",
        "Your playfulness brings such joy!"
    ]
}
}

age_group = 'children' if child_profile.get('age', 30) < 13 else
'elderly'

if child_emotion['primary_emotion'] in verbal_mirrors:
    responses = verbal_mirrors[child_emotion['primary_emotion']]
[age_group]
    return random.choice(responses)

return "I'm here with you."

```

3. Augmented Reality Tea Party with Physical Toy Integration

A. AR Toy Overlay & Interaction System

python

```
class ARTeaPartyManager:
    def __init__(self):
        self.ar_engine = AREngine()
        self.toy_tracker = ToyTracker()
        self.interaction_manager = ARInteractionManager()

    def setup_tea_party_scene(self, toys, child_position):
        """
        Create AR tea party scene with virtual overlays on physical toys
        """
        scene = {
            'table': self.create_virtual_table(child_position),
            'place_settings': [],
            'decorations': [],
            'interactive_elements': [],
            'toy_overlays': {}
        }

        # Create place setting for each toy
        for i, toy in enumerate(toys):
            place_setting = self.create_place_setting(toy, i, len(toys))
            scene['place_settings'].append(place_setting)

            # Add AR overlay to toy (hat, bowtie, etc.)
            overlay = self.create_toy_overlay(toy)
            scene['toy_overlays'][toy['id']] = overlay

            # Add interactive elements near toy
            interactive = self.create_interactive_element(toy, i)
            scene['interactive_elements'].append(interactive)

        # Add decorations
        scene['decorations'] = self.create_decorations(child_position)

        # Add magical effects
        scene['magical_effects'] = self.create_magical_effects()

        return scene

    def create_toy_overlay(self, toy):
        """
        Create AR overlay that makes toy appear to be participating
        """
        overlays = {
            'teddy_bear': {
                'hat': {
                    'type': 'top_hat',
                    'color': '#8B4513', # Brown
                    'animation': 'gentle_bob'
                },
            },
            'bowtie': {
                'type': 'classic_bowtie',
                'color': '#FF6B6B', # Coral
                'animation': 'subtle_pulse'
            },
        },
        'doll': {
```

```

        'tiara': {
            'type': 'princess_tiara',
            'color': '#FFD700', # Gold
            'animation': 'sparkle'
        },
        'necklace': {
            'type': 'pearl_necklace',
            'color': '#FFFFFF',
            'animation': 'gentle_glow'
        }
    },
    'stuffed_animal': {
        'party_hat': {
            'type': 'cone_hat',
            'color': random.choice(['#FF6B6B', '#4ECDC4', '#FFD166']),
            'animation': 'bounce'
        }
    }
}

overlay_config = overlays.get(toy['type'], overlays['stuffed_animal'])

return {
    'assets': overlay_config,
    'position': 'attached_to_toy',
    'tracking': {
        'method': 'feature_tracking',
        'update_rate': 30, # Hz
        'smoothness': 0.8
    },
    'interactive': True
}

def create_interactive_element(self, toy, position_index):
    """
    Create interactive AR elements near each toy
    """
    elements = [
        {
            'type': 'tea_cup',
            'position': 'in_front_of_toy',
            'interactions': ['fill', 'sip', 'spill'],
            'liquid_color': '#D4A76A', # Tea color
            'steam_effect': True,
            'sound_effects': ['pour_tea', 'sip_sound']
        },
        {
            'type': 'plate',
            'position': 'left_of_cup',
            'contents': ['virtual_cookie', 'virtual_cake'],
            'interactions': ['take_cookie', 'share'],
            'animation': 'hover_slightly'
        },
        {
            'type': 'napkin',
            'position': 'right_of_cup',
            'color': '#FFE5CC', # Cream
            'interactions': ['fold', 'wipe'],
            'animation': 'gentle_wave'
        }
    ]

    element = elements[position_index % len(elements)].copy()

```



```

# Personalize based on toy
if toy['personality']['personality'] == 'wise_elderly':
    element['contents'] = ['virtual_biscuit', 'virtual_jam']
elif toy['personality']['personality'] == 'cheerful_energetic':
    element['contents'] = ['virtual_cupcake', 'virtual_lollipop']

return element

def handle_child_interaction(self, child_gesture, ar_element, toy):
    """
    Handle child's interaction with AR elements
    """
    interaction_result = {
        'success': False,
        'reaction': '',
        'visual_feedback': None,
        'sound_effect': None,
        'toy_response': None
    }

    # Determine what child is trying to do
    action = self.interpret_gesture(child_gesture, ar_element)

    if action == 'pour_tea':
        interaction_result.update({
            'success': True,
            'visual_feedback': self.create_pouring_animation(ar_element),
            'sound_effect': 'tea_pouring',
            'toy_response': self.generate_toy_thanks(toy)
        })

    elif action == 'offer_food':
        interaction_result.update({
            'success': True,
            'visual_feedback':
self.create_food_transfer_animation(child_gesture),
            'sound_effect': 'happy_munching',
            'toy_response': self.generate_toy_food_reaction(toy,
ar_element['contents'][0])
        })

    elif action == 'clean_spill':
        interaction_result.update({
            'success': True,
            'visual_feedback': self.create_cleaning_animation(),
            'sound_effect': 'wiping',
            'toy_response': "Oh dear, thank you for cleaning up!"
        })

    return interaction_result

```

B. Toy Conversation & Interaction System

python

```

class ToyConversationSystem:
    def __init__(self):
        self.conversation_topics = self.load_topics()
        self.dialogue_engine = DialogueEngine()
        self.toy_personalities = {}

    def facilitate_toy_conversation(self, toys, child_emotion, context):
        """

```

```

Facilitate conversation between toys during tea party
"""
# Select conversation topic based on context
topic = self.select_conversation_topic(child_emotion, context)

# Assign speaking order
speaking_order = self.determine_speaking_order(toys, child_emotion)

# Generate conversation
conversation = []
for i, toy_id in enumerate(speaking_order):
    toy = next(t for t in toys if t['id'] == toy_id)

    # Generate what this toy says
    dialogue = self.generate_toy_dialogue(
        toy,
        topic,
        conversation[:i], # What's been said so far
        child_emotion
    )

    conversation.append({
        'toy': toy['personality']['name'],
        'dialogue': dialogue,
        'expression': self.generate_toy_expression(toy, dialogue),
        'voice': toy['voice']
    })

    # Occasionally have toys interact with child
    if random.random() < 0.3: # 30% chance
        question_to_child = self.generate_question_for_child(toy, topic)
        conversation.append({
            'toy': toy['personality']['name'],
            'dialogue': question_to_child,
            'type': 'question_to_child',
            'expected_response_type':
self.get_expected_response_type(question_to_child)
        })

    return {
        'topic': topic,
        'conversation': conversation,
        'duration_estimate': len(conversation) * 5, # 5 seconds per line
        'interaction_opportunities':
self.identify_interaction_points(conversation)
    }

def generate_toy_dialogue(self, toy, topic, previous_dialogue,
child_emotion):
    """
    Generate dialogue for a specific toy
    """
    dialogue_templates = {
        'wise_elderly': {
            'tea': [
                "This tea reminds me of the gardens in springtime.",
                "A good cup of tea makes everything better, don't you
think?",
                "I remember when tea parties were grand affairs with music
and dancing."
            ],
            'friendship': [
                "The best part of any party is the friends you share it
with.",

```

```

        "True friendship is like a good cup of tea - warming and
comforting.",
        "We may be different, but we complement each other
beautifully."
    ]
},
'cheerful_energetic': {
    'tea': [
        "Yummy! This tea is delicious! Can I have another cookie
too?",
        "I love the little bubbles in my tea! They're dancing!",
        "Tea parties are the best! So much fun and treats!"
    ],
    'friendship': [
        "You're my best friend! Let's play forever!",
        "Friends make everything more fun!",
        "I'm so happy we're all here together!"
    ]
},
'elegant_gracious': {
    'tea': [
        "The tea service is simply exquisite today.",
        "One must always take time to appreciate the finer things.",
        "This blend has lovely floral notes, don't you think?"
    ],
    'friendship': [
        "It's an honor to be included in such lovely company.",
        "Friendship adds grace to every occasion.",
        "Your kindness makes this gathering truly special."
    ]
}
}

personality = toy['personality']['personality']
templates = dialogue_templates.get(personality,
dialogue_templates['cheerful_energetic'])

# Get relevant templates for topic
topic_templates = templates.get(topic, templates['friendship'])

# Select based on child's emotion
if child_emotion['primary_emotion'] == 'sad':
    # More comforting dialogue
    comforting_templates = [
        "Sometimes a quiet tea with friends is just what we need.",
        "It's okay to have quiet moments. We're here with you.",
        "The warmth of tea and friendship can help any heart."
    ]
    topic_templates = comforting_templates + topic_templates

return random.choice(topic_templates)

def generate_question_for_child(self, toy, topic):
    """
    Generate question for the child to answer
    """
    questions_by_personality = {
        'wise_elderly': [
            "What's your favorite thing about tea parties?",
            "If you could have any kind of tea, what would it be?",
            "What makes a good friend in your opinion?"
        ],
        'cheerful_energetic': [
            "What's the silliest thing you've ever done?",

```

```

        "If you were a cookie, what kind would you be?",
        "What's your favorite game to play?"
    ],
    'elegant_gracious': [
        "What's the most beautiful thing you've seen today?",
        "If you could plan the perfect party, what would it be like?",
        "What makes you feel special?"
    ]
}

personality = toy['personality']['personality']
questions = questions_by_personality.get(personality,
questions_by_personality['cheerful_energetic'])

return random.choice(questions)

```

4. Complete Tea Party Experience Manager

A. Main Tea Party Orchestrator

python

```

class InteractiveTeaPartyOrchestrator:
    def __init__(self, user_profile):
        self.user_profile = user_profile
        self.toy_recognition = ToyRecognitionEngine()
        self.emotion_recognizer = ChildEmotionRecognizer()
        self.emotion_mirroring = EmotionMirroringSystem()
        self.ar_manager = ARTeaPartyManager()
        self.conversation_system = ToyConversationSystem()
        self.marvin_expression = EmotionalExpressionEngine()

        self.session_state = {
            'phase': 'setup',
            'toys_present': [],
            'child_emotion_history': [],
            'tea_party_progress': 0.0,
            'magical_moments': []
        }

    def run_tea_party(self):
        """
        Main tea party orchestration loop
        """
        print("? Welcome to Marvin's Magical Tea Party! ?")

        # Phase 1: Setup
        self.phase_setup()

        # Phase 2: Welcome & Introductions
        self.phase_welcome()

        # Phase 3: Tea Serving
        self.phase_serve_tea()

        # Phase 4: Conversation & Games
        self.phase_conversation_games()

        # Phase 5: Magical Moments
        self.phase_magical_moments()

```

```

        # Phase 6: Farewell
        self.phase_farewell()

def phase_setup(self):
    """
    Child sets up physical toys for tea party
    """
    print("? Setup Phase: Place your toys for the tea party!")

    # Marvin gives instructions
    self.marvin_speak("Welcome! Let's set up our tea party. Place your
favorite toys around you so I can see them all!")
    self.marvin_expression.display_expression('excited', intensity=0.8)

    # Wait for toys to be placed
    toys_placed = False
    while not toys_placed:
        frame = self.get_camera_frame()
        toys = self.toy_recognition.initialize_childs_toys(frame)

        if len(toys) >= 1: # At least one toy
            self.session_state['toys_present'] = toys
            toys_placed = True

            # Celebrate
            self.marvin_speak(f"Wonderful! I see {len(toys)} friends joining
us!")
            self.marvin_expression.display_expression('happy',
intensity=0.9)
            self.play_sound('magical_chime')
        else:
            self.marvin_speak("I don't see any toys yet. Place them in front
of the camera!")
            time.sleep(2)

    # Create AR scene
    ar_scene = self.ar_manager.setup_tea_party_scene(
        toys,
        self.get_child_position()
    )

    self.session_state['ar_scene'] = ar_scene

    # Introduce each toy
    for toy in toys:
        self.introduce_toy(toy)

def introduce_toy(self, toy):
    """
    Have Marvin introduce each toy with personality
    """
    introduction = f"""
    Oh look! It's {toy['personality']['name']}!
    {toy['personality']['name']} is a {toy['type']} who loves
    {random.choice(toy['personality']['favorite_topics'])}.
    """

    self.marvin_speak(introduction)

    # Add AR overlay
    self.ar_manager.add_toy_overlay(toy['id'], toy['personality'])

    # Toy "greet" back
    time.sleep(1)

```

```

toy_greeting = self.generate_toy_greeting(toy)
self.speak_as_toy(toy, toy_greeting)

# Marvin responds
time.sleep(0.5)
self.marvin_speak("So lovely to have you with us!")

def phase_welcome(self):
    """
    Welcome ceremony with magical elements
    """
    print("? Welcome Phase")

    # Marvin gives welcome speech
    welcome_speech = """
    Welcome, everyone, to our magical tea party!
    Today is special because we're all here together.
    Let's begin with a sprinkle of magic...
    """

    self.marvin_speak(welcome_speech)

    # Magical animation
    self.play_animation('sparkle_effect', duration=3)
    self.play_sound('magical_transformation')

    # Each toy gets "magically transformed" with AR
    for toy in self.session_state['toys_present']:
        self.ar_manager.activate_magical_overlay(toy['id'])
        time.sleep(0.5)

    # Check child's reaction
    child_emotion = self.get_child_emotion()
    self.session_state['child_emotion_history'].append(child_emotion)

    # Mirror child's emotion
    mirrored_response = self.emotion_mirroring.generate_mirrored_response(
        child_emotion,
        self.user_profile
    )

    # Apply Marvin's expression
    self.apply_expression_sequence(mirrored_response['expression_sequence'])

    # Speak response
    self.marvin_speak(mirrored_response['verbal_response'])

def phase_serve_tea(self):
    """
    Interactive tea serving with AR
    """
    print("☕ Tea Serving Phase")

    self.marvin_speak("Now, let's serve the tea! Watch carefully...")

    # Demonstrate pouring tea
    self.ar_manager.demo_tea_pouring()

    # Encourage child to pour for toys
    self.marvin_speak("Now you try! Make a pouring motion toward a toy's
cup.")

    # Wait for child interaction
    interaction_detected = False

```

```

start_time = time.time()

while not interaction_detected and (time.time() - start_time) < 60: # 1
minute timeout
    frame = self.get_camera_frame()
    child_gesture = self.analyze_child_gesture(frame)

    if child_gesture.get('type') == 'pouring':
        # Find which toy they're pouring for
        toy_id = self.determine_target_toy(child_gesture)

        if toy_id:
            interaction_detected = True

            # AR feedback
            result = self.ar_manager.handle_pouring_interaction(
                child_gesture,
                toy_id
            )

            # Toy responds
            toy = next(t for t in self.session_state['toys_present'] if
t['id'] == toy_id)
            toy_response = self.generate_toy_thank_you(toy)
            self.speak_as_toy(toy, toy_response)

            # Marvin praises
            self.marvin_speak("Perfect pouring! What a wonderful host
you are!")
            self.marvin_expression.display_expression('proud',
intensity=0.8)

            # Record magical moment
            self.session_state['magical_moments'].append({
                'type': 'tea_poured',
                'toy': toy['personality']['name'],
                'timestamp': time.time()
            })

            time.sleep(0.1)

        if not interaction_detected:
            self.marvin_speak("That's okay! Let me help. Watch as I pour tea for
everyone!")
            self.ar_manager.auto_pour_tea(self.session_state['toys_present'])

def phase_conversation_games(self):
    """
    Interactive conversation and games with toys
    """
    print("? Conversation Phase")

    # Start conversation between toys
    conversation = self.conversation_system.facilitate_toy_conversation(
        self.session_state['toys_present'],
        self.get_child_emotion(),
        {'phase': 'tea_party'}
    )

    self.marvin_speak("Listen! Our friends are having a conversation!")

    # Play out conversation
    for line in conversation['conversation']:
        if line.get('type') == 'question_to_child':

```

```

        # Pause for child to respond
        self.speak_as_toy_by_id(line['toy'], line['dialogue'])

        # Listen for child response
        child_response = self.listen_for_child_response(timeout=10)

        if child_response:
            # Have toy respond to child
            toy_response = self.generate_toy_response_to_child(
                line['toy'],
                child_response,
                line['expected_response_type']
            )
            self.speak_as_toy_by_id(line['toy'], toy_response)
        else:
            # Gentle prompt
            self.marvin_speak(f"What do you think,
{self.user_profile.get('name', 'friend')}?")
        else:
            # Regular conversation line
            self.speak_as_toy_by_id(line['toy'], line['dialogue'])
            time.sleep(line.get('duration', 2))

        # Play tea party games
        self.play_tea_party_games()

    def phase_magical_moments(self):
        """
        Create special magical moments based on child's interactions
        """
        print("✧ Magical Moments Phase")

        # Check for accumulated magical energy
        magical_energy = len(self.session_state['magical_moments'])

        if magical_energy >= 3:
            # Big magical transformation
            self.marvin_speak("Wow! I can feel the magic building up! Everyone,
hold on!")

            # Magical transformation sequence
            self.play_animation('full_transformation', duration=5)
            self.play_sound('orchestral_magical')

            # All toys get enhanced AR effects
            for toy in self.session_state['toys_present']:
                self.ar_manager.apply_magical_transformation(toy['id'])

            # Special celebration
            self.marvin_expression.display_expression('robot_happy_dance',
intensity=1.0)
            self.marvin_speak("Look at that! Our tea party has become truly
magical!")

            # Fireworks
            self.ar_manager.create_fireworks_show()

        elif magical_energy >= 1:
            # Small magical moment
            self.marvin_speak("I feel something magical happening!")

            # Sparkle effect on favorite toy
            favorite_toy = self.determine_favorite_toy()
            if favorite_toy:

```



```

        self.ar_manager.apply_sparkle_effect(favorite_toy['id'])
        self.play_sound('sparkle_twinkle')

        self.marvin_speak(f"Look! {favorite_toy['personality']['name']}
is glowing with happiness!")

    def phase_farewell(self):
        """
        Farewell ceremony
        """
        print("? Farewell Phase")

        # Check child's emotion
        child_emotion = self.get_child_emotion()

        if child_emotion['primary_emotion'] in ['sad', 'reluctant']:
            self.marvin_speak("It looks like you're sad the party is ending. We
can play again soon!")
            self.marvin_expression.display_expression('compassionate',
intensity=0.8)
        else:
            self.marvin_speak("What a wonderful tea party that was!")
            self.marvin_expression.display_expression('happy', intensity=0.9)

        # Each toy says goodbye
        for toy in self.session_state['toys_present']:
            goodbye_message = self.generate_toy_goodbye(toy, child_emotion)
            self.speak_as_toy(toy, goodbye_message)
            time.sleep(1)

        # Final message from Marvin
        final_message = f"""
Thank you for the lovely tea party, {self.user_profile.get('name',
'friend')}!
You were a wonderful host.
Remember, your friends are always here for another party.
Until next time!
"""

        self.marvin_speak(final_message)

        # Wave goodbye animation
        self.marvin_expression.display_expression('content', intensity=0.7)
        self.play_animation('wave_goodbye', duration=3)

        # Save tea party memory
        self.save_tea_party_memory()

```

5. Integration with Main Marvin System

A. Cross-System Memory Integration

python

```

class TeaPartyMemoryIntegration:
    def __init__(self, marvin_core):
        self.marvin_core = marvin_core
        self.memory_bridge = MemoryBridge()

    def save_tea_party_memory(self, session_data):
        """

```

```

Save tea party experience to Marvin's memory system
"""
memory_entry = {
    'type': 'tea_party_experience',
    'timestamp': time.time(),
    'toys_present': session_data['toys_present'],
    'child_engagement': session_data['child_engagement_score'],
    'magical_moments': session_data['magical_moments'],
    'child_emotional_journey': session_data['child_emotion_history'],
    'insights': self.extract_insights(session_data)
}

# Store in chronological dossier
self.marvin_core.memory.add_to_dossier(
    f"Tea Party with {len(session_data['toys_present'])} toys",
    memory_entry
)

# Create semantic beacon for future reference
beacon = {
    'event': 'successful_tea_party',
    'child_preferences': self.extract_child_preferences(session_data),
    'toy_interactions': self.analyze_toy_interactions(session_data),
    'suggested_improvements': self.generate_suggestions(session_data)
}

self.marvin_core.memory.add_semantic_beacon('tea_party', beacon)

# Update user profile
self.update_user_profile_from_tea_party(session_data)

def extract_child_preferences(self, session_data):
    """
    Extract child's preferences from tea party interactions
    """
    preferences = {
        'favorite_toys': [],
        'preferred_interactions': [],
        'engagement_patterns': [],
        'emotional_triggers': []
    }

    # Analyze which toys child interacted with most
    interaction_counts = {}
    for moment in session_data['magical_moments']:
        if 'toy' in moment:
            interaction_counts[moment['toy']] =
interaction_counts.get(moment['toy'], 0) + 1

    if interaction_counts:
        favorite_toy = max(interaction_counts, key=interaction_counts.get)
        preferences['favorite_toys'].append(favorite_toy)

    # Analyze emotional triggers
    emotion_changes =
self.analyze_emotion_changes(session_data['child_emotion_history'])
    preferences['emotional_triggers'] = emotion_changes

    return preferences

def generate_suggestions(self, session_data):
    """
    Generate suggestions for future tea parties
    """

```

```

suggestions = []

# Based on engagement patterns
avg_engagement = np.mean([e['engagement_level'] for e in
session_data['child_emotion_history']])

if avg_engagement < 0.5:
    suggestions.append({
        'type': 'increase_engagement',
        'suggestion': 'Include more interactive games next time',
        'reason': 'Child engagement was below optimal level'
    })

# Based on toy interactions
if len(session_data.get('toy_interactions', [])) < 3:
    suggestions.append({
        'type': 'encourage_interaction',
        'suggestion': 'Prompt child to interact with each toy
individually',
        'reason': 'Limited interaction with toys detected'
    })

return suggestions

```

6. Web Interface & Controls

A. Parent/Child Control Panel

html

```

<div class="tea-party-controls">
  <h2>Marvin's Magical Tea Party</h2>

  <!-- Camera Feed with AR Overlay -->
  <div class="camera-feed">
    <video id="cameraFeed" autoplay></video>
    <canvas id="arOverlay"></canvas>
  </div>

  <!-- Toy Roster -->
  <div class="toy-roster">
    <h3>Party Guests</h3>
    <div id="toyList">
      <!-- Dynamically populated -->
    </div>
  </div>

  <!-- Marvin's Expression -->
  <div class="marvin-expression">
    <canvas id="marvinFace" width="400" height="300"></canvas>
    <div class="marvin-status">
      <span id="marvinMood">Happy</span>
      <span id="marvinEnergy">High</span>
    </div>
  </div>

  <!-- Controls -->
  <div class="party-controls">
    <button onclick="startTeaParty()">Start Tea Party</button>
    <button onclick="addMagicalEffect()">Add Magic</button>
    <button onclick="takeTeaPartyPhoto()">Take Photo</button>
  </div>
</div>

```

```

        <button onclick="endTeaParty()">End Party</button>
    </div>

    <!-- Child's Emotion Display -->
    <div class="child-emotion-display">
        <h3>How You're Feeling</h3>
        <div class="emotion-meter">
            <div class="emotion-bar" id="happinessBar"></div>
            <div class="emotion-bar" id="excitementBar"></div>
            <div class="emotion-bar" id="engagementBar"></div>
        </div>
        <p id="currentEmotion">Detecting emotion...</p>
    </div>

    <!-- Toy Chat -->
    <div class="toy-chat">
        <h3>Party Conversation</h3>
        <div class="chat-log" id="toyChatLog">
            <!-- Conversation appears here -->
        </div>
        <div class="child-response">
            <input type="text" id="childResponse" placeholder="Type your
response...">
            <button onclick="sendResponse()">Send</button>
            <button onclick="speakResponse()">? Speak</button>
        </div>
    </div>
</div>

<script>
// WebSocket connection for real-time updates
const teaPartyWS = new WebSocket('ws://localhost:8765/tea-party');

teaPartyWS.onmessage = function(event) {
    const data = JSON.parse(event.data);

    switch(data.type) {
        case 'toy_detected':
            addToyToRoster(data.toy);
            break;
        case 'marvin_expression':
            updateMarvinExpression(data.expression, data.intensity);
            break;
        case 'toy_speaking':
            addToyToChat(data.toy, data.message);
            break;
        case 'child_emotion_update':
            updateEmotionDisplay(data.emotion);
            break;
        case 'ar_effect':
            showAREffect(data.effect);
            break;
        case 'magical_moment':
            celebrateMagicalMoment(data.moment);
            break;
    }
};

// Camera access
async function startCamera() {
    try {
        const stream = await navigator.mediaDevices.getUserMedia({
            video: { width: 1280, height: 720 }
        });
    }

```

```

        document.getElementById('cameraFeed').srcObject = stream;
    } catch (err) {
        console.error("Camera error: ", err);
    }
}

// Start tea party
function startTeaParty() {
    teaPartyWS.send(JSON.stringify({
        type: 'start_tea_party',
        user_profile: getUserProfile()
    }));

    // Start emotion detection
    startEmotionDetection();
}

// Emotion detection using webcam
function startEmotionDetection() {
    const video = document.getElementById('cameraFeed');
    const canvas = document.createElement('canvas');
    const ctx = canvas.getContext('2d');

    setInterval(() => {
        canvas.width = video.videoWidth;
        canvas.height = video.videoHeight;
        ctx.drawImage(video, 0, 0);

        // Send frame to server for emotion analysis
        const imageData = canvas.toDataURL('image/jpeg', 0.5);
        teaPartyWS.send(JSON.stringify({
            type: 'emotion_frame',
            image: imageData
        }));
    }, 1000); // Send every second
}

// Update emotion display
function updateEmotionDisplay(emotionData) {
    document.getElementById('happinessBar').style.width =
        `${emotionData.happiness * 100}%`;
    document.getElementById('excitementBar').style.width =
        `${emotionData.excitement * 100}%`;
    document.getElementById('engagementBar').style.width =
        `${emotionData.engagement * 100}%`;

    document.getElementById('currentEmotion').textContent =
        `You're feeling ${emotionData.primary_emotion}`;
}

// Initialize
startCamera();
</script>

```

7. Implementation Phases

Phase 1: Basic Toy Recognition (2 weeks)

- Implement toy detection using pre-trained models
- Simple AR overlay system

- Basic emotion detection from camera

Phase 2: Advanced Interaction (3 weeks)

- Gesture recognition for tea pouring
- Toy personality system
- Conversation engine for toys
- Marvin's expression mirroring

Phase 3: Magical Elements (2 weeks)

- AR magical effects
- Progressive magical transformation
- Sound effects and music integration
- Photo capture with AR elements

Phase 4: Polish & Integration (3 weeks)

- Performance optimization for RPi5
- Integration with main Marvin system
- Parent control panel
- Memory system integration

8. Key Features & Benefits

A. For Children:

1. **Imaginative Play:** Physical toys become animated characters
2. **Emotional Intelligence:** Learn to recognize and respond to emotions
3. **Social Skills:** Practice conversation and hosting
4. **Creativity:** Magical transformations encourage creative thinking
5. **Memory Creation:** Magical moments become cherished memories

B. For Parents/Educators:

1. **Developmental Insights:** Track child's emotional and social development
2. **Safe Environment:** Controlled, positive social interactions
3. **Educational Value:** Hidden learning in play
4. **Bonding Opportunity:** Can join tea party remotely

C. Technical Innovations:

1. **Real-Time AR on RPi5:** Optimized for resource constraints
2. **Multi-Modal Emotion Recognition:** Combines facial, body, and vocal cues

3. **Toy Personality AI:** Each toy develops unique characteristics
4. **Progressive Magic System:** Rewards positive interaction with magical moments

Conclusion

This **Augmented Reality Tea Party System** creates a deeply immersive, magical experience that bridges the physical and digital worlds. By:

1. **Transforming physical toys** into animated characters with AR
2. **Responding to child's emotions** with Marvin's expressive face
3. **Creating progressive magical moments** that reward engagement
4. **Fostering social-emotional development** through guided play

We create a therapeutic, educational, and joyful experience that helps children:

- **Develop empathy** through emotional mirroring
- **Practice social skills** in a safe environment
- **Engage in imaginative play** with their favorite toys
- **Build positive memories** of magical interactions

The system maintains the **Peter Protocol's core values** of protection, dignity, and compassionate interaction while adding layers of **magical engagement** that make the tea party experience truly special for children of all ages, including the young at heart.